UNIT 110

Transistor Labs rhythm composer · Owner's manual · version 1.0.4



UNIT 110 is a drum machine you play like the hardware: press its keys, turn its knobs, read its LCD, and open the case to reach the trimmers inside. Six analog-style voices - bass drum, snare, hand clap, closed hat, open hat and cymbal - a global accent, and a 16-step sequencer with 32 patterns and a song mode. It runs as an Audio Unit, VST3 or LV2 plug-in and as a standalone app on macOS, Windows and Linux.

1 System requirements

System	Needs
macOS	**Minimum: macOS 10.13 High Sierra on Intel, macOS 11 Big Sur on Apple silicon** (universal build; runs on macOS 12 Monterey and every later version). A host for AU or VST3 (Ableton Live, Logic Pro, GarageBand, Bitwig, Reaper...), or the standalone app.
Windows	**Minimum: Windows 10, 64-bit** (Windows 11 too). A VST3 host, or the standalone app.
Linux	**Minimum: x86_64** with X11, FreeType and ALSA or JACK. A VST3 or LV2 host (Bitwig, Reaper, Ardour...), or the standalone app.
Screen	The window opens at 1126 x 691 and can be resized from 50 % to 120 %.
CPU / memory	Light: one instance uses a few percent of one core; about 30 MB of memory.

2 Installing

System	File	What it installs
macOS	UNIT110-1.0.4-macOS.pkg	Audio Unit, VST3 and the standalone app. Click Customize to skip any of them.
Windows	UNIT110-1.0.4-Windows-Setup.exe	VST3 (Common Files\VST3) and the standalone app.
Linux	UNIT110-1.0.4-Linux-x86_64.tar.gz	Unpack and run ./install.sh: VST3 to ~/.vst3, LV2 to ~/.lv2, app to ~/.local/bin.

2.1 macOS security warning

The macOS installer is **not notarised** by Apple, so the first time you open it macOS says it "cannot be opened because it is from an unidentified developer" (or that Apple could not check it for malicious software). To install:

1. Double-click the .pkg once and close the warning.
2. Open **System Settings > Privacy & Security**, scroll down to the message about UNIT110 and click **Open Anyway**.
3. Confirm with your password and run the installer.

Alternative: right-click (Control-click) the .pkg in Finder, choose **Open**, then **Open** again.

On Windows, SmartScreen may warn about the installer: click **More info > Run anyway**.

Ableton Live: Settings > Plug-Ins: turn on "Use Audio Units v2" (Mac) and/or VST3, then Rescan. UNIT 110 is under Plug-Ins > Audio Units (or VST3) > Transistor Labs. Drop it on a MIDI track.

3 The front panel

Everything works with the mouse, just like the real keys. Resize the window from its bottom-right corner. Click the TRANSISTOR LABS logo for the About window.

3.0 Opening the case

Click either **screw in the bottom corners** or the **ON-OFF POWER** lettering to open the unit (section 5). Inside, click one of the **screws** or one of the **three big black chips** to close it again (or press Esc).

3.1 Knobs and switch

Control	What it does
SWING	From STRAIGHT (50 %) to SHUFFLE (75 %): every second 16th note comes later. Around 62 % is the classic swing; fully right is a hard, dotted shuffle. The LCD shows the amount (SW) in the TEMPO box. MIDI export swings too.
BALANCE	Drums against metal. Turn left and the hats and cymbal fade; turn right and the bass drum, snare and clap fade. Centre = both full.
ACCENT	How hard accented steps hit: a short swell on the whole machine that also drives it into grit.
VOLUME	Main output level (5 = -14 dB, 10 = +6 dB).
POWER	Click the switch on the top edge. Off = the LCD goes dark and UNIT 110 is silent.
Knobs	Drag up/down. The value shows while you drag; double-click resets.
Tempo	UNIT 110 always follows the tempo of your DAW. In the standalone app, drag up/down on the TEMPO box of the LCD to set it (40-250 BPM); double-click it for 120.

3.2 Keys

Key	What it does
SHIFT	Click it, then the next key does its upper (printed) function. Holding the computer's Shift key while clicking works too. **Double-click** SHIFT to switch the LCD to the P-LOCK screen (section 4.1); double-click again to go back.
A|B|C|D	Steps through the 4 banks of 8 patterns (A1-D8). With SHIFT, or in WRITE STEP: I-II, switches between steps 1-8 and 9-16.
1 - 8	Selects pattern 1-8 of the current bank. While playing, the new pattern starts at the next step 1 (the LCD shows NEXT).
START / STOP	Runs and stops the sequencer while the DAW is stopped, at the DAW's tempo. When the DAW plays, UNIT 110 follows it.
Drum keys	Play the drum. ACCENT arms an accent for the next drum you hit (the key stays down).
SHIFT + drum	**Mute** the drum on/off. Its row on the LCD shows M, the name blinks and its steps go faint.
Option + drum	**Solo** the drum on/off (Alt on Windows / Linux). Its row shows S and the other drums go faint. Several drums can be soloed.
Right-click drum	**Own output** (plug-in only): sends the drum to its own output instead of the main mix (section 6).

3.3 SHIFT functions

SHIFT +	Function	What it does
1	PLAY SONG	Plays the song (the list of patterns built in WRITE SONG), one pattern per bar, locked to the song position.
2	PLAY PATTERN	Plays the selected pattern over and over (the default).
3	WRITE SONG	Number keys add patterns of the current bank to the end of the song. O.HIHAT (ENTER) adds the playing pattern; C.HIHAT (D.C.) takes back the last measure. Up to 32 measures.
4	WRITE STEP	Press ACCENT or a drum key to choose the row (marked on the LCD), then number keys 1-8 switch its steps on and off. A|B|C|D (I-II) switches to steps 9-16.
5	WRITE TAP	Record live: drum keys and MIDI notes played while the sequencer runs go into the nearest step. ACCENT before a hit, or velocity above 100, also sets the accent.
6	12 / 16	Pattern length 12 (shuffle, triplets) or 16 steps.
7	PATTERN CLEAR	Clears the selected pattern.
8	MEASURE RESET	Clears the song.
drum key	MUTE	Mutes / unmutes that drum (see 3.2).

4 The LCD

The upper part is the pattern: 7 rows (ACCENT, BASS DR, SNARE, CLAP, C.HIHAT, O.HIHAT, CYMBAL) by 16 steps. The playing step is shown reversed in the number row. Steps beyond the length are greyed out.

Click a step to switch it on or off, or drag along a row to paint. **Right-click** the LCD for: export the pattern, all patterns in a row, every pattern as its own file, or the song as MIDI; copy the pattern to another slot; clear the pattern or the song. MIDI files use 16th notes on channel 10, accented steps at velocity 127.

The lower boxes show the MODE (and short messages), the PATTERN (bank letter + number, NEXT for a queued one) and the TEMPO, the length, the swing (SW) and whether the host (HOST) or UNIT 110 itself (RUN) is playing. M and S after a drum's name mean muted and soloed.

Copy / Paste: right-click the LCD, choose **Copy pattern**, select another pattern with the number keys, right-click again and choose **Paste pattern**. The steps and the parameter locks are copied. **Copy pattern to** sends it straight to any slot.

4.1 Parameter locks (P-LOCK screen)

A parameter lock gives one step of one drum its own setting: a higher tuned snare on step 7, a longer open hat on the last step. **Double-click SHIFT**: the LCD shows the locks of one drum for one setting, as a bar per step, and the SHIFT key stays down.

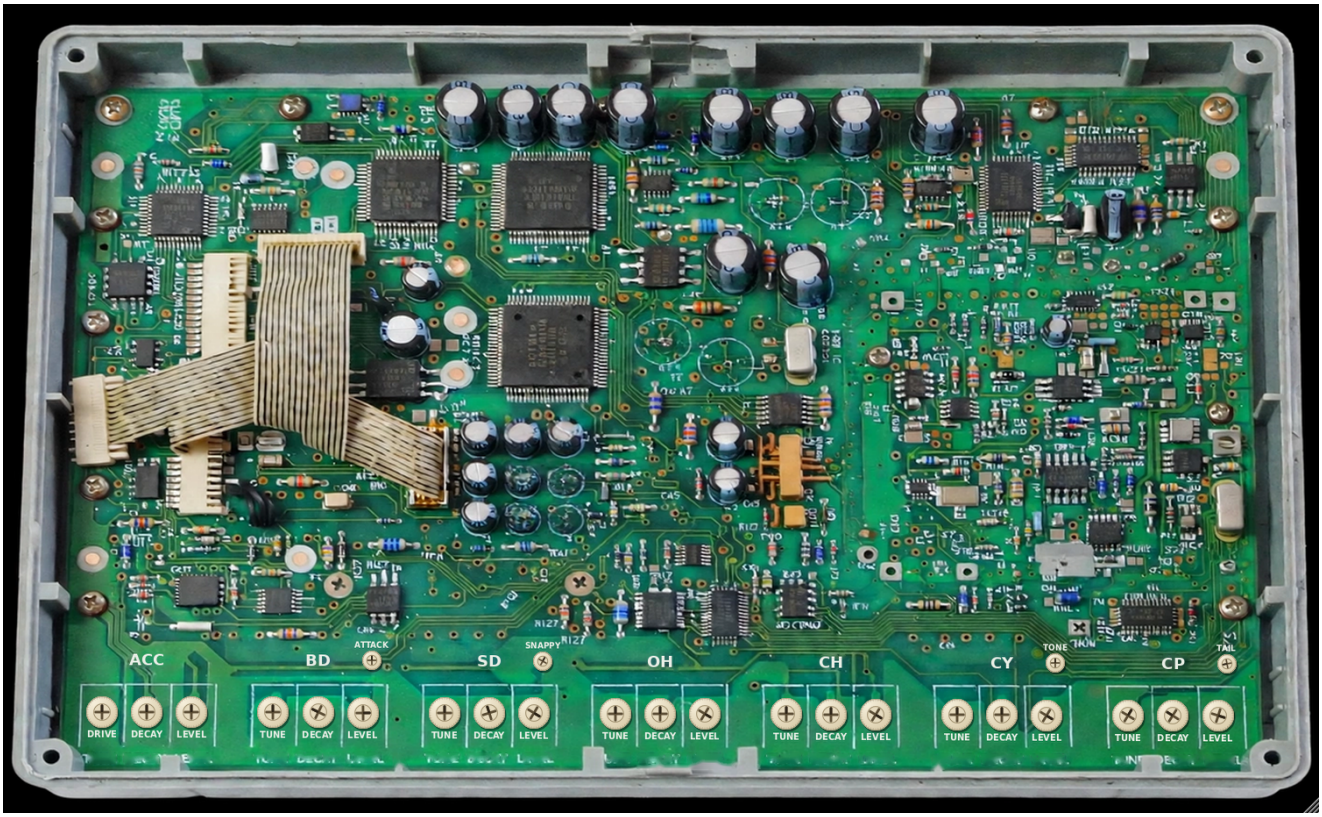
Choose the drum by pressing its drum key (or click the DRUM box). Click the SETTING box to step through TUNE, DECAY, LEVEL and the drum's extra (ATTACK, SNAPPY, TAIL or TONE). **Click or drag** in a step's column to set its value: top = maximum, bottom = minimum (TUNE is centred on 0). **Double-click** (or Alt-click) a column to remove the lock: that step then follows the trimmer again. A mark under a column shows where the drum plays. Right-click for clearing a lane, a drum or the whole pattern.



A lock holds until that drum's next hit. Locks are saved with each pattern and copied with it. Steps with locks show a small dot in the normal pattern view.

5 Inside the case

Click either **screw in the bottom corners** of the front panel to open the unit. The circuit board has a service trimmer for every sound setting. Drag a trimmer up/down; double-click resets it. Click a corner screw on the board, right-click, or press Esc to close the case.



Trimmer	What it does
TUNE	Pitch, ±12 semitones. On the hats and cymbal it moves the whole metallic cluster.
DECAY	Length of the sound. Middle is the classic timing; fully left a quarter as long, fully right four times as long. On the bass drum it is the ring of its resonant filter; on the snare the length of the noise; on the clap the last burst and the room tail.
LEVEL	The drum's level, -inf to +6 dB.

Trimmer	What it does
ACC: DRIVE	How much an accent pushes the machine into grit.
ACC: DECAY	How long the accent swell lasts.
ACC: LEVEL	Accent amount (the same setting as the ACCENT knob on the front).
ATTACK (BD)	How far the kick's pitch drops at the start: 50 % = one octave (110 Hz down to 55 Hz), 0 % = no drop, 100 % = from three times the pitch.
SNAPPY (SD)	Noise snap against the 235 Hz tone. 60 % is the original balance.
TONE (CY)	Bright attack shimmer (right) against the darker ringing body (left).
TAIL (CP)	Level of the clap's room tail after the burst of hits. 15 % is the original.

6 MIDI and outputs

Drum	MIDI notes (C1 = 36)
Bass drum	35, 36
Snare	37, 38, 40
Hand clap	39
Closed hat	42, 44
Open hat	46
Cymbal	49, 51, 52, 53, 55, 57

Velocities up to 100 scale the level; above 100 also fires the accent.

Outputs (plug-in): main stereo plus one mono output per drum. Right-click a drum key and choose **Own output** to take that drum out of the main mix and send it to its own output (in Live: make an audio track, Audio From = the UNIT 110 track, pick the drum's output, Monitor In). The standalone app has only the main output, so there is no right-click menu there.

All 32 patterns, the song, the mode and every knob and trimmer are saved with your project.

7 Specifications

Voices	6: bass drum (55 Hz resonant filter with an octave pitch drop), snare (235 Hz tone + 3550 Hz noise), hand clap (3 + 1 noise bursts and a room tail at 1140 Hz), closed hat, open hat, cymbal; plus global accent
Sequencer	16 steps (12 or 16), 7 rows, 32 patterns (4 banks x 8), song of up to 32 measures, swing 50-75 %, parameter locks, host sync, own clock, step and tap writing
Outputs	Main stereo + 6 mono drum outputs (plug-in)
Formats	AU, VST3 (macOS); VST3 (Windows); VST3, LV2 (Linux); standalone app on all three
Sample rates	Any host rate (44.1-192 kHz)

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